

Faculty of Computer & Information Sciences
Ain Shams University
Course Code: 4XX
Course Name: Computer Vision
Exam: (MidTerm) 24/3/2018
Year: (2nd term) 4th undergraduate (SC)



Instructors: Prof. Dr. Sayed Fadel
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Offering Dept: Scientific Computing
Academic year: 2017-2018
Duration: 75 Minutes
Total Grade: 30 Marks

Question 2:

- A)** Many Computer Vision algorithms such as **SIFT** (Scale-Invariant Feature Transform) seek to detect and analyze features at multiple scales of analysis. (6 Marks).
- i) Explain briefly what is “scale-space”. **Lec 5 slide 9**
 - ii) For what operation is the **Difference-of-Gaussian** (DoG) used in different scale levels? Show its advantage over **Laplacian-of-Gaussian** (LoG). **Lec 5 slides 10, 11, 12**
 - iii) How SIFT achieves scale and rotation invariance. **Lec 5 slides 18 - 24**
- B)** Describe the main stages of the **RANSAC** algorithm in general case. **Lec 6 slide 22**
What kind of a model fitting is computed by RANSAC in case of panoramic image stitching? **Affine transformation model (Lec6 slide 6)**
- C)** Explain the advantage of using the **Mean-shift** algorithm over **K-means** clustering for image segmentation. **Lec2 slide 49**
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With Best Regards,
Prof. Dr. Sayed Fadel
Dr. Dina Khattab